Event-Aware Representation Learning for Neutral Multi-Source News Summarization

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1 The Project Idea (Overview)

The Concept: A neutral multi-source news generation system whose internal representation is explicitly organised around the *event* being reported, rather than around the publisher or framing style of the article.

The Goal: To train the model's latent space so it groups differently-framed reports of the same event together, while keeping reports of different events apart – even when those events come from the same publisher.

The Deliverable: A summarisation model augmented with an event-aware representation-learning objective, plus two optional extensions (label-efficient pretraining and an explicit shared/unique information split), evaluated on how much event information is retained versus how much publisher-specific information leaks through.

2 The Problem (Why This Matters)

NeuS already generates neutral summaries from multiple news articles of varying political leanings, but shows that generation models can hallucinate politically biased content, not just factually inaccurate content (Lee et al., 2022). Separately, work on ideology prediction shows that news encoders can easily latch onto source-specific characteristics – a publisher's stylistic "fingerprint" – rather than the content a task actually needs (Baly et al., 2020). If a summarisation encoder does the same, two failure modes follow: (1) **framing leakage**, where a summary's tone still shifts with which publishers were sampled even though the text looks neutral, and (2) **neutrality by deletion**, where the model appears neutral only by dropping contested details rather than reconciling them. Standard neutral-summarisation training supervises only the output text, so nothing forces the encoder to organise its representation around event identity rather than source identity. We need a training signal that does this explicitly.

3 Our Proposed Approach

We combine two existing ideas differently: Baly et al. (2020) use a triplet objective to strip publisher identity out of a representation while preserving political ideology; Lee et al. (2022) generate neutral summaries from same-event articles without constraining what the representation organises around. We flip the triplet idea: preserve *event identity* while removing *source information*, and use that representation to drive generation.

3.1 Core: Event-Aware Triplet Learning + Neutral Generation

The backbone is a pretrained sequence-to-sequence model (BART). Articles describing the same event are each passed through the same encoder, combined into a single event-level representation, and decoded into a neutral summary, trained with the usual next-token generation objective against a human-written target.

Alongside this we add a triplet objective: for an anchor article (say, a Reuters report on event A), the positive is a different publisher's report of the same event A, and the negative is a Reuters report of a different, topically similar event B. The encoder is trained so the anchor sits closer to the positive than the negative by a margin. Since anchor and negative always share a publisher, the only way to satisfy this consistently is for the model to stop relying on publisher style and instead organise its

representation around the event itself. The full training signal is just the generation loss plus a weighted version of this triplet loss.

Evaluation covers two axes: *event retention*, via DiverseSumm-style QA evaluation checking whether a generated summary can answer questions drawn from multiple source articles (Huang et al., 2024); and *publisher leakage*, via a small classifier trained on frozen representations to predict the source publisher (lower accuracy is better), plus a check for whether summary tone shifts with the publisher mix. The target is high event-QA performance with low publisher recoverability, compared to a model trained with the generation objective alone.

3.2 Extension 1: Cross-Source Pretraining

Following the JEPA approach to self-supervised learning, which predicts one view's latent representation from another rather than reconstructing raw content (Assran et al., 2023), we pretrain by predicting one publisher's representation of an event from another publisher's representation of the same event, using unlabelled same-event pairs. The question: after this pretraining, can the decoder be fine-tuned on only 1–25% of labelled neutral summaries and still match a fully-supervised model?

3.3 Extension 2: Shared vs. Unique Factorisation

Following FactorCL's shared/unique split for multimodal representation learning (Liang et al., 2023), we separate each article's representation into information shared across same-event reports and information unique to that report, generating neutral summaries mainly from the shared component. This generalises beyond "remove publisher X" to any paired-text setting.

4 Literature & Latest Papers (State of the Art)

- **NeuS** (Lee et al., 2022) (NAACL 2022): defines the neutral multi-source summarization task and the same-event article clusters our generation objective trains on.
- **We Can Detect Your Bias** (Baly et al., 2020) (EMNLP 2020): source of the triplet-loss technique we repurpose to preserve event identity instead of ideology.
- **I-JEPA** (Assran et al., 2023) (CVPR 2023): the self-supervised, representation-prediction approach behind our label-efficient pretraining extension.
- **FactorCL** (Liang et al., 2023) (NeurIPS 2023): the shared/unique information factorisation underlying our second extension.
- **DiverseSumm** (Huang et al., 2024) (NAACL 2024): the QA-based evaluation we use to confirm event information is retained, not deleted.